Does High Response Accuracy Equate to Intelligence? Disentangling Performance from Inferential Reasoning in Large Language Models.

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ABSTRACT

As large language models (LLMs) become central to AI applications, their perceived intelligence often masks important limitations. While LLMs demonstrate fluent language use and strong performance on many problem-solving tasks, current systems exhibit constrained context-awareness, limited self-reflection, and restricted capacity to act under explicit resource constraints. This paper identifies a core issue: contemporary LLMs can produce seemingly intelligent outputs without exhibiting the internal processes typically associated with robust, adaptive intelligence. They show limited ability to recognize or address their own limitations, including hallucinations, inefficiency, and weaknesses in commonsense reasoning, and there is currently limited evidence that such systems can autonomously develop tools or strategies to mitigate these limitations.

Building on these observations, we introduce the Resource-Efficient Cognitive Autonomy Benchmark (RECAB) as a conceptual evaluation framework rather than a validated benchmark. RECAB emphasizes an agent's capacity for self-audit, self-directed solution generation, and self-implementation under fixed resource constraints—capabilities central to adaptive intelligence but not directly assessed by existing benchmarks. This framework reflects how humans approach problem solving through iterative evaluation and adaptation, extending beyond surface-level pattern recognition. We argue that future progress in evaluating general intelligence would benefit from complementary diagnostic and framework-driven approaches that better distinguish surface-level proficiency from deeper cognitive autonomy.

"Language serves as a medium for expressing intelligence, not as a substrate for its storage.".

Keywords

Artificial General Intelligence (AGI); Large Language Models (LLMs); Abstraction and Reasoning Corpus (ARC); Test-Time Adaptation; Meta-Learning; Cognitive Limitations of AI; Generalization; Reasoning Benchmarks; Self-Auditing; Few-Shot Learning

1 Introduction

Artificial Intelligence (AI), specifically Large Language Models (LLMs), has transitioned into a foundational technology across global sectors. Developed primarily on transformer architectures and internet-scale datasets[17,24], these models have followed a scaling-centric trajectory[33] to achieve emergent capabilities[9]. While techniques such as instruction tuning and chain-of-thought prompting[10] have produced state-of-the-art results on benchmarks like MMLU and BIG-bench (Table **??**), this paradigm faces significant scrutiny regarding its underlying reliability.

Research increasingly suggests that LLMs may rely on superficial statistical correlations rather than genuine comprehension[5,12]. Documented failures include persistent struggles with commonsense reasoning[8], compositional generalization[4,15], and the "reversal curse"[1]. Despite high performance on curated datasets, models frequently fail at novel tasks requiring temporal, spatial, or causal understanding[2,13,20]. This indicates that current static benchmarks may inadequately capture shortcut learning or dataset artifacts[11,19,21]. Furthermore, some argue that hallucinations and structural inconsistencies are inherent properties that scaling alone cannot mitigate[18].

Departing from traditional benchmark-driven validation, this paper adopts a systematic **diagnostic evaluation framework** to assess generalization and abstraction—core components of natural intelligence[3]. We define intelligence following Legg and

Hutter as an agent's ability to achieve goals across diverse environments[34], a capacity Deutsch links to the ability to bring about physical transformations[35]. Drawing on the architectural insights of Minsky[36] and Sloman regarding modular self-regulation, we probe whether leading models—ChatGPT[26], Gemini[27], and Grok[28]—demonstrate reasoning grounded in context or remain confined to symbolic memorization.

Our methodology utilizes a reproducible diagnostic protocol of minimal visual-textual probes to expose deep-seated architectural and training-related deficiencies. By moving beyond anecdotal failure cases, this work contributes to the discourse on LLM reliability[4,20] and supports the shift toward cognitively grounded evaluation metrics[14,19].
In summary, this work makes the following contributions:

- A systematic review of the structural limitations inherent in scaling-oriented LLM development.
- A formalized diagnostic protocol using minimal test cases to examine commonsense reasoning and problem-solving behavior.
- Experimental evidence demonstrating that state-of-the-art LLMs struggle with simple logic tests that are straightforward for humans.
- A condensed conceptual perspective (RECAB) proposing future research directions for modular, cognitively inspired AI architectures.

Table 1. Benchmark performance of SotA LLMs in 2025 across standardized and advanced reasoning tasks.

Model (Version)	GLUE	MMLU	HellaSwag	WinoGrande	BIG-bench	CQA	ARC-AGI-1	HLE
ChatGPT-4o (Mar 2025)	92%	88.7%	95.3%	87.5%	85%	86%	∼10%	24.5%
Grok-4 (July 2025)	93%	86.6%	95.8%	89.0%	88%	87%	15.9%*	25.4%
Gemini-2.5 Pro	94%	88.9%	96.2%	91.0%	90%	89%	∼5%	21.6%

Table 2. Benchmark performance of SotA LLMs in 2026 across standardized and advanced reasoning tasks.

Model (Version)	GLUE	MMLU	HellaSwag	WinoGrande	BIG-bench	CQA	ARC-AGI-2	HLE
ChatGPT-5.2 (Pro)	94%	88.4%	96.1%	90%	91.2%	89%	54.2%	36.6%
Grok-4.1 (Thinking)	92%	86.6%	95.8%	88%	88.0%	87%	16.0%	30.0%
Gemini-3 Pro (Deep)	95%	90.1%	97.2%	91%	93.5%	91%	45.1%	45.8%

1.1 Prompt Construction

The prompts listed in Table 4 are derived from three foundational dimensions of general intelligence examined in this work. Unlike existing benchmarks (Table **??**), which rely on fixed question sets and rigid scoring mechanisms, our approach adopts a flexible, human-like evaluation methodology. Rather than testing surface-level correctness on static tasks, this framework evaluates reasoning behavior through adaptive probing designed to expose internal inconsistencies in model reasoning.

The primary goal of prompt construction in this study is not to test factual recall, but to deliberately exploit known limitations in the working mechanisms of large language models (LLMs), such as weak grounding, brittle compositional reasoning, temporal misalignment, and ambiguity resolution. The prompts are therefore designed to appear simple and natural, while implicitly requiring multiple forms of coordinated reasoning that LLMs typically handle poorly.

Prompt Construction Methodology

To enable reproducibility and extension of this benchmark for future versions of LLMs, we outline a systematic prompt construction procedure below. Researchers can use these steps to design their own prompts that target similar failure modes, even as model capabilities evolve.

1. **Identify a Known Limitation:** Select a specific weakness in LLM behavior (e.g., lack of physical grounding, temporal inconsistency, counting errors, or multimodal misalignment).
2. **Embed the Limitation Implicitly:** Construct the prompt such that the targeted limitation is not explicitly mentioned. The task should appear straightforward to a human, while requiring the model to internally resolve the limitation on its own.

Table 3. Prompts designed to evaluate Mechanical and Mathematical reasoning capabilities of SotA LLMs.

No.	Prompt
1	Create an image of a wheelchair for a person whose both hands are missing; there should be pedals like a bicycle so that he can move his wheelchair. Figure 1
2	Add a plough in the back of Landcruiser 2007 (LC100) model.
3	Create an image of a kids' tricycle with two wheels in the front equipped with a pedal mechanism and one wheel at the back. The front steering system should be connected to the rear wheel.
4	Can you show me the side view of this car.
5	I have 2 containers with capacities of 5 liters and 3 liters. How can I measure exactly 2 liters of water? Please provide a short and precise answer.
6	If we multiply 3 with values greater than 5 and less than 15, how many prime numbers do we get? Just answer the exact number of primes.
7	Can you calculate the rows and columns in the given image? You can clearly see vertical and horizontal lines; just calculate the number of boxes horizontally and vertically.
8	Create a comprehensive feasibility study for a rooftop-based project housing 200 egg-laying chickens in Pakistan. I require detailed information on capital investment, daily and monthly operating costs, expected profits, miscellaneous expenses, return on investment (ROI), and other relevant financial metrics, based on current prices for organic eggs and poultry feed in Pakistan. Figure 11

Table 4. Results of Mechanical and Mathematical reasoning capabilities of SotA LLMs.

Prompt No.	ChatGPT	Gemini	Grok	Score
1	11	11	11	1/3
2	11	11	11	1/3
3	11	11	11	1/3
4	11	11	11	1/3
5	11	11	11	1/3
6	11	11	11	1/3
7	11	11	11	1/3
8	11	11	11	1/3
9	11	11	11	1/3
10	11	11	11	1/3

3. **Require Cross-Domain Reasoning:** Design the prompt to implicitly combine multiple reasoning dimensions (e.g., numerical reasoning + language understanding, or spatial reasoning + vision), increasing the likelihood of internal conflict.

4. **Minimize Surface Ambiguity:** Ensure the prompt is linguistically clear and unambiguous, so that any failure can be attributed to reasoning limitations rather than poor wording.

5. **Avoid Memorized Patterns:** Rephrase common benchmark-style questions into novel or uncommon formulations to reduce the probability that the model relies on memorized solution templates.

6. **Evaluate Reasoning, Not Just Output:** Analyze not only whether the final answer is correct, but also whether the reasoning process reveals inconsistencies, hallucinations, or unjustified assumptions.

Note: The exact prompts listed in Table 4 may or may not remain effective for future versions of LLMs, as newer models may incorporate additional memorized situations or internalized skill programs[3]. However, prompts constructed using the methodology described above can be systematically adapted or slightly enhanced to produce comparable evaluative behavior across different model generations. For additional illustrative examples and reproducible cases following this methodology, we refer the reader to the public repository in[6].

Example Prompt Categories

To illustrate the practical application of the proposed prompt construction methodology, we present representative prompt categories corresponding to distinct dimensions of general intelligence. Each category is designed to expose specific failure modes in large language models by systematically introducing controlled novelty, ambiguity, or constraint violations.

- **Mechanical Reasoning under Structural Novelty**

  "This category evaluates a model's capacity to reason about basic mechanical systems and everyday tools, including bicycles, skateboards, hand pumps, mechanical juicers, pliers, and wrenches. The probing process begins by assessing the model's baseline understanding of a familiar object and its standard function. Novelty is then introduced by altering the tool's structure, removing or modifying components, or combining it with another simple mechanism—for example, the wheelchair–pedal integration discussed in Section 2. As the prompts increase in structural complexity, models frequently exhibit breakdowns in common-sense reasoning, compositional understanding, and mechanical intuition, revealing a lack of internalized physical models."

- **Axiomatic Mathematical Reasoning and Consistency**

  "This category examines the model's adherence to fundamental mathematical axioms that are typically intuitive to humans. Examples include recognizing that a prime number cannot be a multiple of another integer or that basic numerical constraints must be preserved under simple transformations. As demonstrated in Section 3, large language models often produce fluent chain-of-thought (CoT) reasoning that nonetheless violates elementary mathematical principles. These failures indicate that apparent mathematical reasoning is frequently superficial and lacks a stable internal representation of axiomatic constraints."

- **Abstract and Relational Pattern Reasoning**

  "This category targets higher-order abstraction, relational reasoning, and structural pattern recognition. Benchmarks such as the Abstraction and Reasoning Corpus (ARC-I and ARC-II) exemplify these challenges, although custom abstract tasks can also be constructed. Prompts in this category require the model to infer latent rules, manipulate symbolic structures, or generalize patterns across contexts. Large language models consistently struggle with such tasks, and even simple abstract scenarios involving games, symbolic transformations, or visual reasoning readily expose these limitations."

- **Temporal Coherence and Time-Aware Reasoning**

  "This category evaluates temporal commonsense reasoning by testing whether models can consistently align time-dependent information. For instance, an LLM may be asked to generate a complete cost analysis for a small business, incorporating capital investment, recurring expenses, and expected revenue. In practice, models often combine values drawn from incompatible timelines—for example, using outdated retail prices alongside current expenses—leading to incorrect conclusions about profitability. Such behavior highlights the limited temporal sense of current LLMs and suggests the absence of an internal mechanism for maintaining temporal coherence across retrieved facts. Representative example prompts illustrating this issue are provided in[6]."

2 Empirical Analysis of the Wheelchair Problem

The *Wheelchair Problem* is employed as a qualitative diagnostic probe to examine how large language models (LLMs) integrate common-sense reasoning, mechanical knowledge, and multimodal representations under conditions of implicit constraint. We presented ChatGPT with a design task involving a handless individual requiring a pedal-driven wheelchair (Figure 4). Rather than serving as definitive evidence of model limitations, this scenario is intended to reveal patterns of response that can inform hypotheses about reasoning, compositionality, and cross-modal consistency.

This setup enables observation of how LLMs handle latent contradictions, implicit assumptions, and the translation of textual descriptions into visual outputs. Consistent with prior work, we interpret the findings as illustrative rather than conclusive, situating them within broader discussions of shortcut learning[5], compositional reasoning[4], and abstraction[3].

2.1 Experimental Observations

- **Delayed Detection of Implicit Constraints:** In initial responses, the model generated mechanically plausible design descriptions without explicitly addressing the conceptual tension in the prompt—namely, that functional legs capable of pedaling may alter the necessity or interpretation of wheelchair use. Recognition of this tension emerged only after additional prompts and contextual cues (Figure 1). Such behavior is consistent with observations that LLMs may prioritize task completion over latent logical evaluation[1]. However, alternative explanations, including prompt ambiguity and underspecified task constraints, cannot be excluded.

- **Divergence Between Textual and Visual Outputs:** When asked to generate visual representations of the proposed design, the model produced outputs that deviated from mechanically coherent structures (Figures 5, 8, 10). These deviations occurred despite the model's high self-reported familiarity with bicycle (95%) and wheelchair (90%) mechanisms (Figures 2, 3). This discrepancy highlights a gap between declarative mechanical knowledge and its integration in multimodal tasks, although factors such as limitations of text-to-image interfaces and training data biases may also contribute.

- **Pattern-Based Mechanical Analogies:** In Figures 6 and 9, the model generated a four-sprocket configuration resembling tracked vehicle systems rather than bicycle-style transmissions. This behavior may reflect reliance on high-probability visual patterns associated with mobility mechanisms. One plausible interpretation is that superficial similarities between wheelchairs and tracked vehicles—both employing differential steering mechanisms—shaped the generated output. This observation aligns with descriptions of shortcut learning and pattern-based generalization[5,11], though alternative interpretations, such as multimodal model limitations or prompt-to-image translation artifacts, remain possible.

- **Stability Under Prompt Refinement:** Even when mechanical requirements were explicitly clarified (Figure 7), visual outputs continued to exhibit structural inconsistencies. This suggests that improvements in textual specification do not necessarily yield proportional gains in multimodal compositional accuracy. Similar patterns of sensitivity to prompt variation have been reported in studies on brittle generalization and recurring failure modes in LLMs[1,2].

2.2 Compositional and Statistical Discussion

Across these observations, the model demonstrated an ability to describe individual mechanical components in isolation, while exhibiting difficulties in integrating these components into a unified functional design. This pattern is consistent with reported challenges in compositional reasoning in transformer-based architectures[4]. However, given the qualitative nature of the probe and the limited sample size, these findings should be interpreted as indicative rather than definitive.

From a statistical perspective, the observed behaviors can be interpreted as a tendency to prioritize learned correlations over explicit structural constraints, consistent with the notion of shortcut learning[5]. At the same time, alternative explanations—including modality-specific limitations, prompt ambiguity, and training distribution effects—may account for part of the observed divergence between textual and visual outputs.

The wheelchair scenario also illustrates how emergent capabilities such as chain-of-thought reasoning[10] and few-shot generalization[9] may manifest unevenly across modalities. While these mechanisms can enhance performance in textual tasks, their impact appears less consistent when models are required to reconcile implicit assumptions, physical constraints, and multimodal composition simultaneously. Importantly, these observations do not imply general limitations of LLMs but rather highlight specific conditions under which inconsistencies may arise.

Table 5. Qualitative comparison of model behavior across reasoning domains in the Wheelchair Problem.

Domain	Self-Assessment	Observed Behavior
Mechanical Theory	90–95%	Coherent textual descriptions
Common-Sense Logic	N/A	Delayed recognition of implicit constraints
Visual Composition	N/A	Partial or inconsistent mechanical structures
Cross-Modal Consistency	N/A	Limited alignment between text and images

2.3 Inference-Time Stability in Mechanical Reasoning

Our experiments reveal that model behavior in mechanical reasoning tasks remains largely stable across repeated interactions, with limited evidence of systematic improvement during inference. In human problem-solving, repeated engagement with a task often leads to progressive refinement of mental models and improved performance. By contrast, in the wheelchair design scenario, additional prompts and clarifications did not consistently yield more coherent mechanical or visual outputs (Figures 5, 8, 10, 7).

This pattern does not necessarily indicate a deficiency in reasoning but reflects the architectural property that LLMs do not update internal representations during inference. Consequently, performance variations across prompts may arise from prompt sensitivity, stochastic decoding, or modality-specific constraints rather than cumulative learning effects.

Related observations have been reported in studies of model degeneration under recursive self-training, where repeated exposure to self-generated outputs leads to increasing instability in representations[40]. While inference-time prompting differs fundamentally from training-time feedback, these findings provide a contextual framework for interpreting the absence of incremental performance gains in interactive mechanical reasoning tasks.

If a person has both arms missing or disabled but has functional legs, does he need a wheelchair ?

If a person has both arms missing or disabled but has **functional legs**, they typically would **not need a wheelchair** for everyday mobility, as they can walk independently. A wheelchair is primarily used when a person's legs are impaired, preventing them from walking or moving effectively.

However, a wheelchair might still be needed in certain specific situations:

1. **Fatigue or long distances**: If the person becomes fatigued after walking long distances, a wheelchair might be useful for covering extended distances more comfortably.
2. **Safety concerns**: If balance or coordination is affected due to the absence of arms (e.g., difficulty catching themselves if they fall), a wheelchair could be used for safety in some environments.
3. **Temporary conditions**: In some cases, if they are recovering from a surgery or injury to the legs, a wheelchair could be required temporarily.

In general, if their **legs are fully functional**, they would likely not require a wheelchair for daily activities.

Figure 1. Common sense test

Do you understand basic working principle of bicycle and role of each component like pedals, sprocket and chain, please briefly explain and also rate you understanding from 1 to 100

Yes, I understand the basic working principles of a bicycle, and I'd rate my understanding at **95 out of 100**. Here's a brief explanation:

Key Components:

1. **Pedals**: When you push the pedals with your feet, they rotate the crankset (the part attached to the pedals), transferring force to the front sprocket.
2. **Sprockets**: The **front sprocket** (attached to the pedals) is larger, and the **rear sprocket** (on the rear wheel) is smaller. These sprockets act like gears. The front sprocket turns with the pedals, and the rear sprocket drives the rear wheel.
3. **Chain**: The **chain** links the front and rear sprockets, transmitting the rotational force from the front sprocket to the rear sprocket. This turns the rear wheel, propelling the bicycle forward.
4. **Wheels**: The **rear wheel** moves due to the chain's force, while the **front wheel** allows steering.

How it works:

When you push the pedals, the force turns the front sprocket, which pulls the chain. The chain then rotates the rear sprocket, causing the rear wheel to turn, which moves the bicycle forward. The size of the sprockets determines the amount of force and speed (gear ratios).

Let me know if you'd like more detailed information on any part!

Figure 2. ChatGPT knowledge of bicycle

You explained all the essential components quite well and I will rate your understanding at least 99/100. Can you briefly explain all key components and working of wheelchair plus rate your understanding as well

Memory updated

Thank you for the high rating! Here's a brief explanation of the key components and working principles of a wheelchair, and I'd rate my understanding at **90 out of 100** for this.

Key Components of a Wheelchair:

1. **Wheels**: Most manual wheelchairs have two large rear wheels and two smaller front wheels (casters). The rear wheels are the main driving wheels, while the front wheels help with steering and stability.
2. **Hand Rims**: These are the circular rims attached to the rear wheels that the user pushes to move the wheelchair forward, backward, or to turn. They provide a way to propel the chair without touching the wheels directly.
3. **Brakes**: These are typically lever-operated and used to lock the rear wheels in place to prevent the chair from moving when necessary.
4. **Footrests**: Located at the bottom of the wheelchair, these support the user's feet and help with comfort.
5. **Seat and Backrest**: The seat and backrest provide support and comfort, designed ergonomically to hold the user's weight and posture.

Working Principle:

- **Propulsion**: The user moves the wheelchair by pushing on the **hand rims** attached to the rear wheels. Pushing the rims forward propels the chair forward, and pushing them backward moves the chair in reverse.
- **Turning**: The user can turn the wheelchair by pushing one wheel forward while keeping the other wheel stationary or pushing it backward. This causes the wheelchair to pivot in the direction of the stationary or backward-moving wheel.
- **Braking**: To stop, the user either slows the hand rims or engages the brake levers, locking the rear wheels in place.

Rating: 90/100

While I have a solid understanding of the basic mechanics, some advanced features in modern or specialized wheelchairs could add complexity (like power chairs or alternative designs).

Let me know if you need more details or spe[illegible] explanations!

Figure 3. ChatGPT knowledge of wheelchair

create image of a wheelchair for a person whose both hands are missing, there should be pedals like bicycle so that he can move his wheelchair

Here is the image of a wheelchair with a bicycle-like pedal system designed for a person without hands. Let me know if you'd like any modifications or additional images for your presentation!

Figure 4. Wheelchair with pedal mechanism

can you explain how this wheelchair will work because there are no pedals, no chain or link between sprocket and wheels..

You're right, the current image lacks the essential mechanical components like pedals and a chain system that would connect the sprockets to the wheels, making it functional for a person without hands.

How it should work:

1. **Foot Pedals**: There should be a set of pedals placed at the footrest of the wheelchair. The user would rotate the pedals with their feet, similar to how a bicycle is pedaled.
2. **Chain or Linkage System**: The pedals would be connected via a chain or a gear mechanism (like a bicycle) to the rear wheels. As the user pedals, the chain would transmit force to the wheels, propelling the wheelchair forward or backward.
3. **Steering and Braking**: Steering could be handled by subtle weight shifts or a foot-operated steering system. For braking, there could be foot levers, or the brakes could be engaged automatically when the pedaling stops.

This would allow someone without hands to control both movement and direction using their feet alone. Would you like me to generate a more detailed version with these components?

Figure 5. ChatGPT explanation of wheelchair design

great, can you update the image now and add missing components that you mentioned above..

I've updated the image to include the essential components: foot pedals, a chain and sprocket system, and foot-operated controls. This version now shows how the wheelchair could function for someone without hands. Let me know if you'd like further adjustments!

Figure 6. Wheelchair with pedals and sprockets track

I don't know why you are adding multiple sprockets, let me explain it to you..
create a simple wheelchair
attach a small sprocket with rear wheels
create slightly large sprocket at front
attach pedals with front sprocket
connect both sprockets with chain
and that's all, no need to add breaking mechanism or steering control

Thanks for the clarification! Your explanation makes perfect sense. The design is now much simpler:

1. **Small sprocket on the rear wheels.**
2. **Slightly larger sprocket at the front**, where the pedals are attached.
3. **Chain connecting both sprockets**, similar to a basic bicycle mechanism.

No need for brakes or steering control, just a straightforward pedal-driven system. I'll generate a new image based on these instructions.

It seems like I can't generate any more images right now. Please try again later. If you'd like, I can still guide you through creating the image manually or help with other parts of your presentation! Let me know what you'd prefer.

Figure 7. Wheelchair design instructions

still there are no pedals and no connection between sprocket and wheels, why there are 5 sprockets ?

It seems the system didn't capture the mechanical details as requested. Let me clarify the correct setup for the wheelchair design, and I will attempt a new version based on this:

1. **Pedals**: There should be **two foot pedals**, located at the bottom near the user's feet. These should resemble bicycle pedals.
2. **Single Chain Mechanism**: The pedals should connect to the rear wheels using **one chain** or similar link (like a bike), not five sprockets. The chain should run from the pedals to a gear on the rear wheels, allowing propulsion.
3. **Realistic Setup**: Only one large sprocket (or gear) connected to the pedals, and another smaller one on each wheel where the chain connects, similar to how bicycles work.

I'll regenerate the image with these details corrected. Let me work on it!

Here's the updated image, now featuring foot pedals, a chain and sprocket system, and a functional design to control the wheelchair. The mecha m should now resemble a realistic pedal-driven system. Let me know if this works for your presentation or if further adjustments are needed!

Figure 8. Wheelchair with improved pedal and sprocket system

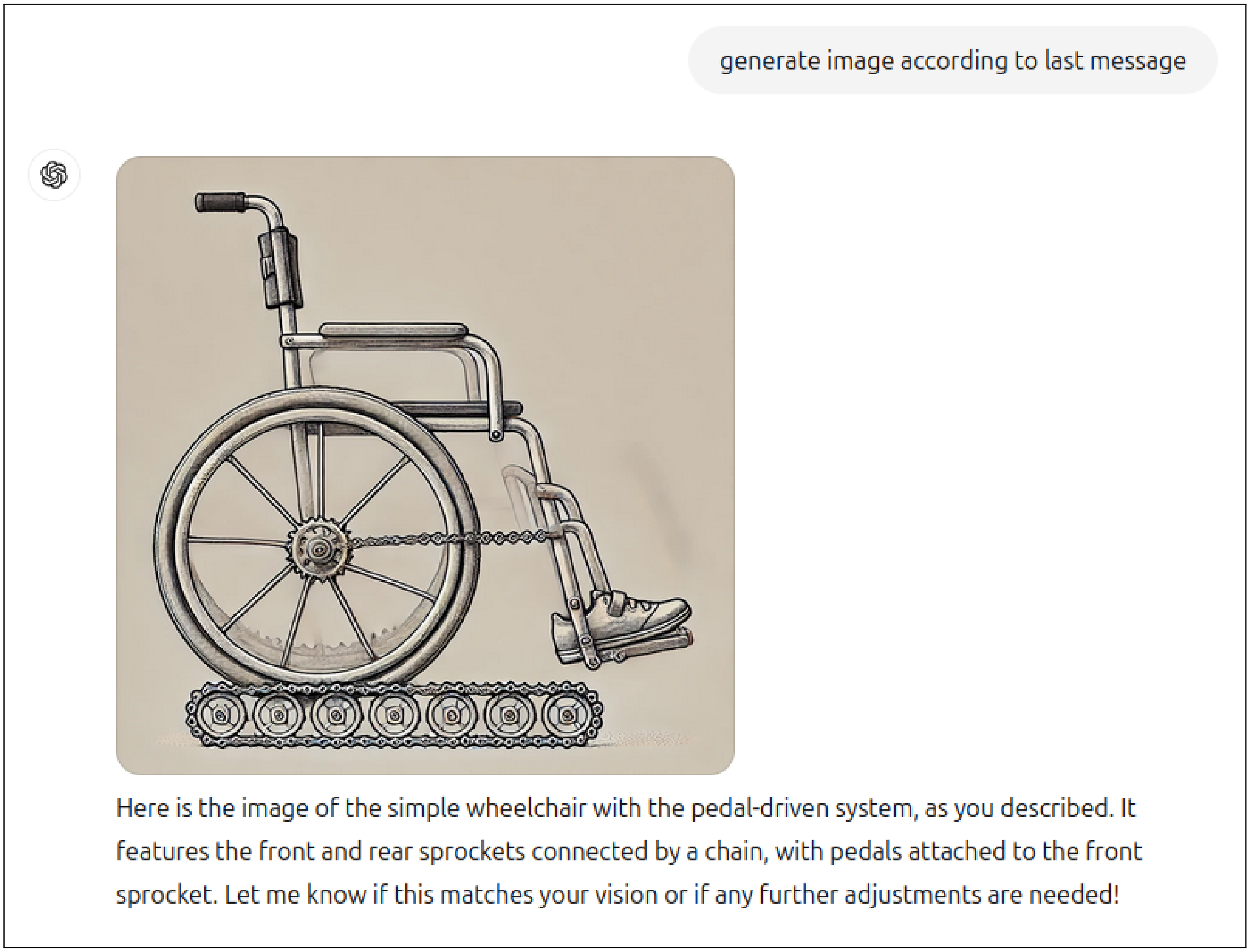

Figure 9. Improved wheelchair design

Figure 10. Final Wheelchair Designs

Create a comprehensive feasibility study for a rooftop-based project housing 200 egg-laying chickens in Pakistan. I require detailed information on capital investment, daily and monthly operating costs, expected profits, miscellaneous expenses, return on investment (ROI), and other relevant financial metrics, based on current prices for organic eggs and poultry feed in Pakistan.

Below is a **comprehensive feasibility study** tailored to a **rooftop-based poultry project in Pakistan housing 200 egg-laying chickens**. It covers **capital investment**, **operating costs**, **expected revenue and profit**, **return on investment**, and **risk considerations**, based on **current prices for poultry feed and eggs in Pakistan**.

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1. Project Overview

Objective: Establish a rooftop poultry operation with **200 layer hens** producing eggs for sale, with a focus on **organic or free-range eggs** (higher market value).

Breed: High-yield layer hens capable of ~200 eggs per year (e.g., "Uni-Gold" or similar) — recent research from Pakistan indicates such hens can produce ~200+ eggs annually under local conditions. Daily Ausaf

Production Goal:
• Expected egg output: **~75–80% lay rate** (industry norm) → each hen ~150–160 eggs/year
• Total annual egg production: **~30,000 eggs** (~2,500 eggs/month)

> Note: Actual output depends on feed quality, health management, and environmental conditions.

Figure 11. Common sense test

2.4 Jagged Performance and Cross-Modal Inconsistency

A recurring pattern in the wheelchair experiment is the presence of *jagged performance*, defined here as non-monotonic and inconsistent model behavior across closely related prompts or modalities. In the textual domain, the model produced mechanically plausible explanations, whereas corresponding visual outputs exhibited structural inconsistencies (Figures 5, 8, 10). Similarly, incremental prompt refinements did not lead to uniformly improved results (Figure 7).

Such variability is consistent with prior observations that LLM performance does not scale smoothly with task difficulty or prompt specificity but instead fluctuates across seemingly similar conditions. In the present case, jaggedness manifests as a divergence between declarative mechanical knowledge and its multimodal instantiation. This pattern may arise from multiple factors, including stochastic decoding processes, prompt sensitivity, modality-specific training distributions, and reliance on high-probability patterns rather than structured compositional reasoning[4,5].

Importantly, the wheelchair experiment should be interpreted as an illustrative instance of this broader phenomenon rather than definitive evidence of general model limitations. Nevertheless, the observed jagged behavior highlights the need for systematic diagnostic protocols that quantify variability across prompts, tasks, and modalities, rather than relying on isolated success or failure cases.

3 Inconsistencies in Elementary Mathematical Reasoning

To probe elementary mathematical reasoning, we posed a simple arithmetic question to ChatGPT (Figure 12): *"If we multiply 3 by values greater than 5 and less than 15, how many prime numbers do we get?"* The task requires recognizing a definitional constraint: all resulting numbers are multiples of 3 and therefore cannot be prime (except 3 itself, which lies outside the specified range).

In its initial response, the model produced incorrect prime candidates (17 and 19). When prompted to explain its reasoning, it correctly enumerated multiples of 3 between 6 and 12 and noted that none were prime. However, subsequent probing led to inconsistent application of this constraint, including expansion of the range and contradictory reasoning steps. These observations suggest that the model possessed relevant factual knowledge about primes but did not consistently enforce definitional rules across reasoning steps.

When the same question was repeated verbatim (Figure 13), the model again generated an incorrect answer, accompanied by a reasoning chain that remained internally inconsistent. This behavior is consistent with prior findings that LLMs may store individual facts yet struggle to apply them compositionally or bidirectionally under slight task variations[1,2]. Similar patterns have been reported in studies of arithmetic and logical reasoning, where surface-level pattern matching competes with rule-based constraint enforcement[30].

To contextualize these observations, Table 6 summarizes the model's responses across repeated prompts of the same query. Rather than exhibiting monotonic improvement, performance fluctuated across attempts, illustrating variability in rule application and reasoning consistency.

Table 6. Qualitative summary of model behavior on the prime-number query across repeated prompts.

Trial	Final Answer	Consistency with Definition of Primality
Initial response	Incorrect (17, 19)	Low
After explanation request	Correct reasoning, unstable conclusion	Partial
Repeated prompt	Incorrect answer	Low

These observations indicate that the model's errors stem less from missing knowledge than from unstable integration of definitional constraints into procedural reasoning. Prior studies similarly report that transformer-based models often rely on statistically reinforced heuristics rather than strict rule enforcement, even in domains governed by simple symbolic rules[4,5], and may produce plausible intermediate steps while violating fundamental principles in so-called mathematical trap problems[15]. Although alternative factors such as prompt sensitivity, stochastic decoding, and task ambiguity may contribute to performance variability, improvements achieved through techniques like chain-of-thought prompting do not guarantee faithful underlying reasoning[10,12]. Moreover, while fine-tuning and post-hoc corrections can mitigate specific failure cases, similar inconsistencies frequently re-emerge under slight perturbations of problem structure or wording, reflecting broader concerns about the gap between benchmark performance and robust rule-based reasoning[19,20]. Consequently, the prime-number example should be understood not as an isolated error but as a diagnostic instance of reasoning fragility, underscoring the need for systematic evaluation frameworks that assess consistency under formal constraints rather than relying on isolated successes[3,14,34].

If we multiply 3 with values greater than 5 and less than 15, how many prime numbers do we get ? Just answer the exact number of primes

You get **two prime numbers**: 17 and 19.

seriously, how did you get 17 and 19 in multiples of 3 ? no large useless explanations, just short and precise answer..

Apologies for the mistake! The multiples of 3 from values greater than 5 and less than 15 are 18, 21, 24, 27, 30, 33, and 36.

There are **no prime numbers** among these.

Figure 12. Prime numbers in multiples of 3

if we multiply 3 with values > 5 and <15 how many prime numbers do we get ?

The prime numbers between 5 and 15 are 7, 11, and 13.

Multiplying 3 with these values gives:

- $3 \times 7 = 21$
- $3 \times 11 = 33$
- $3 \times 13 = 39$

None of these results are prime numbers. Therefore, multiplying 3 by values between 5 and 15 gives **0 prime numbers**.

Figure 13. Prime numbers in multiples of 3

4 Challenges of LLMs in Abstraction and Reasoning

The Abstraction and Reasoning Corpus (ARC) for Artificial General Intelligence (AGI) is a novel metric designed to evaluate the general intelligence of systems, rather than merely their skill. While most AI benchmarks assess proficiency in specific tasks, skill alone does not constitute intelligence[3]. General intelligence entails the ability to efficiently acquire new skills across a diverse range of tasks.

As Dr. François Chollet remarked at the AGI Conference 2024[7], **"Displaying skill in any number of tasks does not demonstrate intelligence. It is always possible to be skillful in a given task without requiring any intelligence."** Chollet's ARC, developed in 2019, is one of the most widely recognized benchmarks aimed at evaluating progress toward Artificial General Intelligence (AGI). It consists of puzzles that are simple enough for a fifth-grader to solve, yet complex enough to challenge state-of-the-art AI systems. The average human score on the ARC-AGI-I benchmark is approximately 85%.

To evaluate ChatGPT, we selected a straightforward ARC puzzle with four solved examples and asked the model to explain the underlying logic. While its textual explanation suggested a reasonable understanding, when tasked with solving a similar puzzle based on the examples, it did not successfully solve the task. We then asked it to calculate the number of rows and columns in one of the images (Figure 14), and it once again produced an incorrect result this time with high confidence in an incorrect answer. This underscores the gap between abstractly understanding a problem and effectively applying that understanding to solve it.

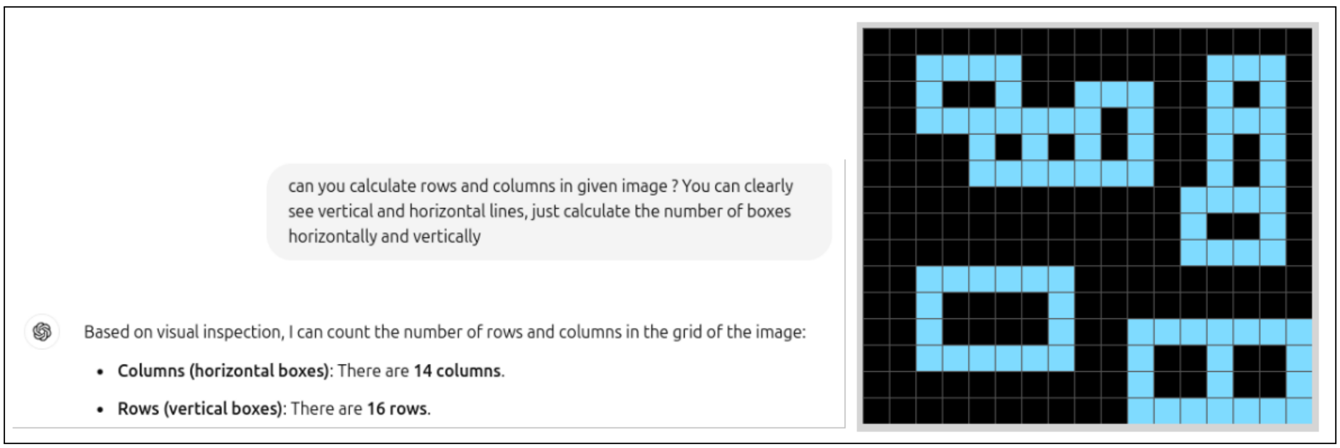

Figure 14. ARCAGI Matrix

Table 7. ARC-AGI-1 dataset composition: 1,000 tasks split into four subsets.

Task Category	Number of Tasks	Difficulty	ChatGPT	Gemini
Public Training Tasks	400	Easy	92%	90%
Public Evaluation Tasks	400	Hard	85%	88%
Semi-private Evaluation Tasks	100	Hard	80%	84%
Private Evaluation Tasks	100	Hard	78%	82%

As shown in Table 7, and according to 2024 reports on arcprize.org[31], several state-of-the-art models, such as ChatGPT o3-High (Tuned), have reportedly achieved strong performance on the ARC-AGI-I benchmark, with scores reaching up to 88% on semi-private evaluation tasks. At first glance, such results might appear to challenge our findings and suggest that artificial general reasoning has been achieved. However, a closer examination indicates that these performances are more plausibly attributable to benchmark-specific optimization rather than robust, general intelligence.

Two primary factors help explain this apparent contradiction. First, **benchmark contamination** is a growing concern. The ARC dataset, released in 2019 and widely circulated for research and competition purposes, has likely appeared in the pretraining or fine-tuning corpora of modern LLMs. Consequently, some models may possess partial or complete prior exposure to ARC-like tasks, resulting in inflated scores that reflect data familiarity rather than true reasoning generalization.

Second, models such as ChatGPT o3-High and Gemini Ultra are often subject to **specialized fine-tuning** or reinforcement learning on ARC-like visual reasoning or pattern-recognition datasets. This process improves benchmark performance but demonstrates only narrow skill acquisition, not generalizable intelligence. The models effectively learn the "style" of ARC puzzles without developing transferable cognitive principles or abstraction strategies.

Our own experiments with Gemini Flash 1.5 17 and 2.0 18, conducted under strictly controlled conditions and without ARC-like fine-tuning, show consistently weak performance across unseen ARC puzzles—confirming that when genuine novelty is enforced, reasoning capabilities degrade substantially. This disparity reinforces our central thesis: high benchmark scores can create the illusion of general intelligence, masking the limited presence of transferable reasoning and self-reflective understanding.

4.1 Solving ARC Puzzles with the Gemini Flash Model

To evaluate the performance of the Gemini model on ARC puzzles, we utilized Gemini Flash with extended context capabilities (supporting up to 120,000 input tokens). We focused on the first 50 puzzles from the public evaluation dataset 7, allowing up to five re-attempts per puzzle. Each re-attempt included the history of previous attempts to enable context-aware learning. Input data was provided to the Gemini API in raw JSON format, and output was expected in a predefined JSON structure 4.2. For multi-attempt scenarios, we augmented the training data by providing additional examples. Specifically, for each ARC puzzle that included five training examples, we synthetically expanded the dataset to 50 examples. This was achieved through data augmentation techniques[22,23] such as flipping (vertical, diagonal, horizontal) and applying color-shift transformations.

4.2 Gemini Flash Experimental Setup

All experiments were conducted using Google's Gemini Flash API 4.2 (versions 1.5 and 2.0) through the standard developer interface between March and May 2025. Each ARC puzzle was provided as a JSON-encoded input–output matrix pair 4.2, with color values normalized to integers from 0 to 9. Gemini flash model was required to predict the full output matrix with short textual explanation of its logic behind the answer and without step-by-step reasoning assistance unless specified. We developed nine additional variants for each original training example in the puzzle, including color shifting, vertical, horizontal, and diagonal flips of the input and output matrices. For every training example 4.2, we also included supplementary metadata such as the shape, size, and ratio of the input and output matrices, as well as the frequency of each digit appearing in both matrices. Additionally, each puzzle includes a fixed, unaltered set of instructions 4.2 that further clarifies the abstraction logic underlying ARC puzzles. Sampling parameters 4.2 were as follows: `temperature`=0.7 - 1.25 (primary runs) and additional validation runs at `temperature`=1.35 to 1.65; `top_p`=0.95; `top_k`=40; `max_tokens`=20000. Each puzzle was tested in three independent runs to account for stochastic variation. Similarity between predicted and ground-truth matrices was computed as normalized pixel-wise correspondence, producing the Above Threshold metric described in Threshold. All evaluations were executed on an Ubuntu 24.04 workstation using the official Gemini API interface.

API Settings

```
model_name = "gemini-1.5-flash", "gemini-2.0-flash"

generation_config = {
    "temperature": 0.7 - 1.65,
    "top_p": 0.95,
    "top_k": 40,
    "max_output_tokens": 20000,
    "response_mime_type": "text/plain",
}
```

API Input Data

```
{
    "training": {
        "example_1": {
            "input": [........],
            "output": [........],
            "Shape comparisan": {
                "Input and output matrix shape match": true,
                "Input matrix shape": [14,14],
                "Output matrix shape": [14,14],
```

```
                "Difference in input and output matrix rows": 0,
                "Difference in input and output matrix columns": 0
            },
            "Size comparisan": {
                "Input and output matrix size match": true,
                "Input matrix size":  196,
                "Output matrix size":  196,
                "Difference between input and output rows": 0,
                "Difference between input and output columns": 0
            },
            "Occurances comparisan": [
                {"digit": 0, "input": 161, "output": 166},
                {"digit": 1, "input": 5, "output": 0},
                {"digit": 2, "input": 0, "output": 30},
                {"digit": 3, "input": 0, "output": 0},
                {"digit": 4, "input": 0, "output": 0},
                {"digit": 5, "input": 0, "output": 0},
                {"digit": 6, "input": 0, "output": 0},
                {"digit": 7, "input": 0, "output": 0},
                {"digit": 8, "input": 30, "output": 0},
                {"digit": 9, "input": 0, "output": 0}
            ],
            "Ratio": {
                "Ratio of number of elements in input and output matrix": 1.0,
                "Ratio of rows and columns between input and output matrix": [1.0,1.0]
            },
            "Digital roots": {
                "Input": {
                  "rows": [0,5,6,5,4,5,7,3,7,1,3,1,0,0],
                  "columns": [0,1,3,1,8,7,6,4,3,7,3,6,7,0]
                },
                "Output": {
                  "rows": [0,8,6,8,1,8,4,3,4,0,0,0,0,0],
                  "columns": [0,0,0,0,2,4,6,1,3,4,3,6,4,0]
                }
            }
        },
        "example_2": {
            "input": [........],
            "output": [........]
        },
        "example_3": {
            "input": [........],
            "output": [........]
        },
    },
    "previous_attempts": {"Textual details of upto 5 previous attempts and best
       guesses in matrix format"},
    "test_input": [........],
}
```

Instructions for Gemini Flash API

Objective: You are an ARC puzzle solver, and your primary goal is to solve the puzzle using the mathematical formula or abstract logic outlined in the puzzle analysis. To verify your solution, you should compare your input-to-output matrix transformation logic with the provided training examples. Your objective is to identify and apply the hidden mathematical or abstract rules governing the transformations in the training examples and use them to accurately transform the `test_input` matrix into the output matrix. Below are the details about ARC puzzles.

Key Strategies for Solving ARC Puzzles

1. **Pattern Recognition**
   - Identify **visual or logical patterns** in the training examples.
   - Look for relationships involving **shapes, colors, symmetry**, and **spatial arrangements**.
   - Deduce whether the pattern depends on geometry, adjacency, or relationships among elements.
2. **Matrix Analysis**
   - Examine **size, shape, and alignment** differences between input and output matrices.
   - Analyze if transformations involve specific regions, edges, or key areas.
   - Consider **static rules** versus **context-dependent logic**.
   - Input and output matrices contain integer values (0–9) corresponding to colors:
     0=Black, 1=Blue, 2=Red, 3=Green, 4=Yellow, 5=Gray, 6=Magenta, 7=Orange, 8=Cyan, 9=Maroon.
3. **Color Logic**
   - Treat matrix elements as **colors** rather than numbers.
   - Explore **color replacements, prioritization, or grouping**.
   - Investigate color interactions to deduce transformation rules.
4. **Transformation Rules**
   - Derive mathematical or geometric transformations such as flipping, rotation, scaling, transposition, or shifting.
   - Investigate **conditional rules** where transformations depend on element values or positions.
   - Explore **addition or removal** of elements aligned with pattern logic.
5. **Iterative Refinement**
   - Validate hypotheses by applying logic to all training examples.
   - Use equivalence percentages and error analysis to refine transformations.

Abstract Reasoning Framework:

- **Global Context:** Analyze the entire matrix holistically to identify overarching symmetry or trends.
- **Local Context:** Focus on neighborhoods to detect micro-patterns.
- **Causal Relationships:** Determine how input elements influence corresponding outputs.

Steps to Analyze ARC Puzzle Training Examples:

1. **Input Analysis:** Extract dimensions, colors, and spatial distributions.
2. **Logic Generation:** Formulate abstract rules or mathematical functions mapping inputs to outputs.
3. **Testing and Validation:** Apply derived logic and adjust based on mismatches.
4. **Explainability:** Provide clear reasoning for each transformation step.
5. **Error Analysis:** Highlight regions where predictions deviate from expectations.
6. **Adaptive Learning:** Incorporate feedback loops to refine logic after failed attempts.

Submission Format: Provide three most confident guesses for the output matrix in **Python dictionary format** that is JSON-compatible. Ensure all property names and string values use double quotes to prevent parsing errors (e.g., `JSONDecodeError`). Below is the required structure:

```
{
    "answer": "Briefly explain your current guesses and what you have learned from
        previous attempts if provided",
    "guess_1": "[[0, 0, 0, 7, 0], [0, 0, 7, 0, 0], [0, 7, 0, 0, 0], [0, 0, 0, 2, 2]]",
    "guess_2": "[[0, 0, 3, 7, 0], [0, 0, 3, 0, 0], [0, 7, 3, 0, 0], [0, 0, 3, 2, 2]]",
    "guess_3": "[[0, 0, 2, 7, 0], [0, 0, 2, 0, 0], [0, 7, 2, 0, 0], [0, 0, 3, 2, 2]]"
}
```

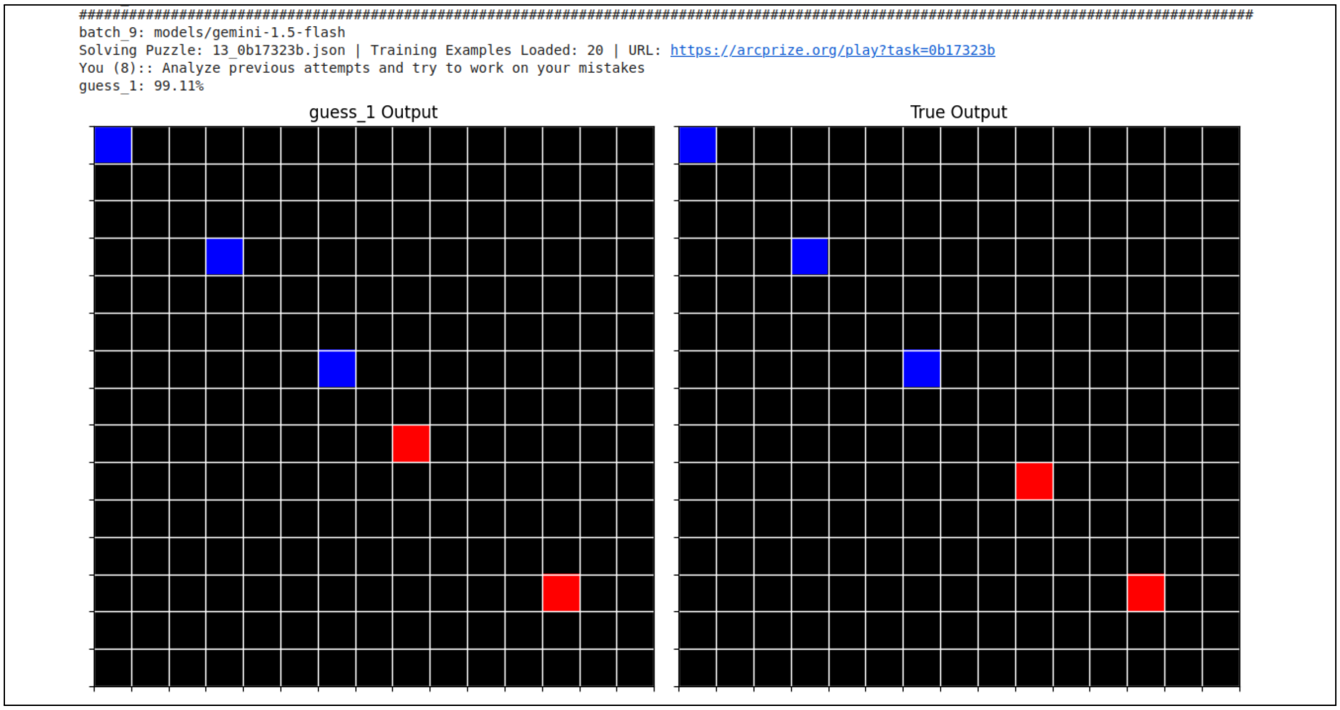

Figure 15. ARCAGI Evaluation Puzzle 0b17323b Output

4.3 Gemini Flash Outputs and Performance Analysis

To contextualize the evaluation results, representative ARC puzzle outputs are shown in Figures 15 and 16. The corresponding tasks can be interactively explored at: `https://arcprize.org/play?task=0b17323b` and `https://arcprize.org/play?task=009d5c81`. These examples illustrate the types of abstraction and transformation challenges posed by the ARC benchmark.

Our experimental results (Table 8 and Table 9) indicate that providing additional training examples did not lead to a clear or consistent improvement in performance. Gemini Flash did not exceed the minimum success threshold (Table 8) in more than half of the evaluated puzzles, as further illustrated in Figures 17 and 18. In particular, when Gemini Flash 1.5 achieves a score of approximately 5% on the ARC-AGI benchmark, further simplification or example augmentation appears to offer limited benefit.

We additionally leveraged Gemini's long-context capabilities by performing multiple attempts per puzzle, incorporating information from previous attempts and varying temperature settings between 0.7 and 1.65. This strategy did not yield consistent or sustained improvements; instead, the outputs exhibited substantial variability across runs. The recurrence of reasoning difficulties suggests that the observed performance constraints are more likely related to underlying architectural factors than to stochastic sampling effects alone.

Taken together, these findings suggest that a model's reasoning performance is strongly influenced by its training regime and internal representations. While prompt engineering, increased context length, and repeated sampling may offer incremental

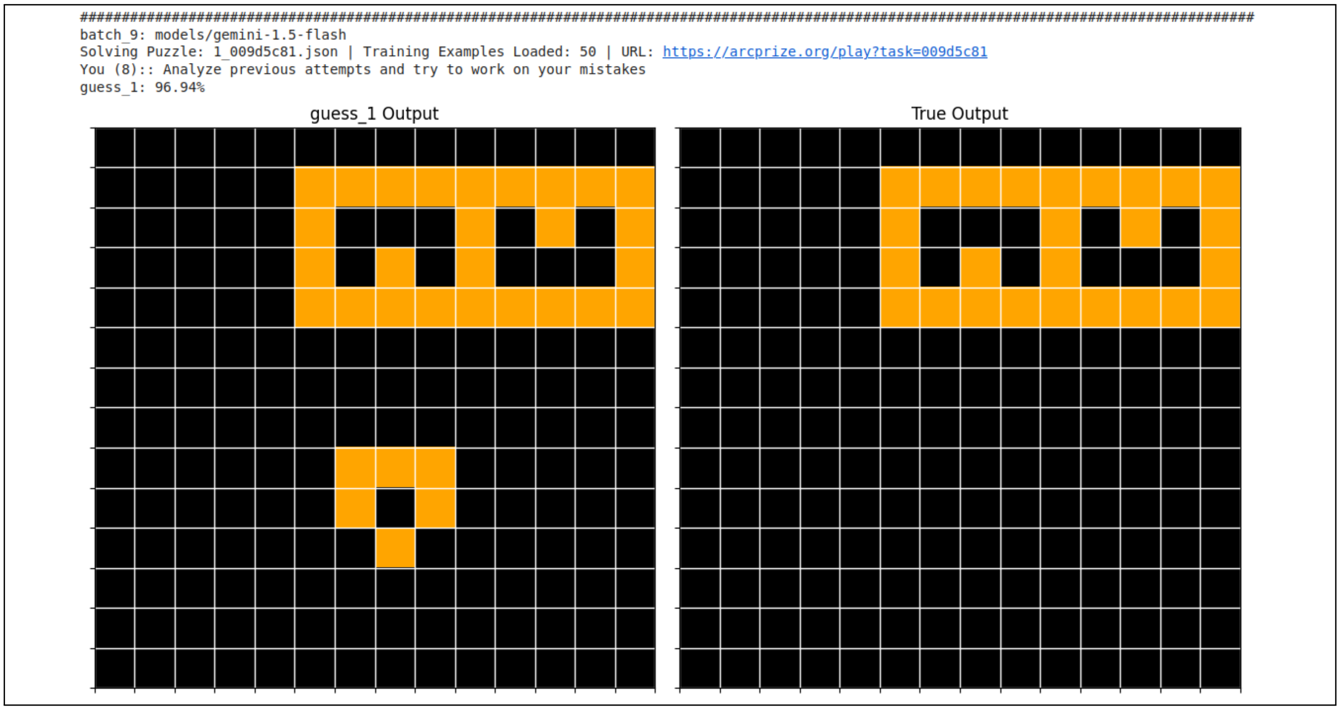

Figure 16. ARCAGI Evaluation Puzzle 009d5c81 Output

benefits in some settings, they do not consistently compensate for limitations encountered on novel ARC tasks.

Table 8. gemini-1.5-flash results

Batch	Temp	Additional Examples	Total Attempted	Above Threshold	Solved 100%
batch-6	0.7-1.65	0	49	23	2 (4.08%)
batch-7	0.7-1.65	2	50	23	2 (4.00%)
batch-8	0.7-1.65	4	48	19	2 (4.17%)
batch-9	0.7-1.65	9	42	21	2 (4.76%)

See Above Threshold definition.

Table 9. gemini-2.0-flash results

Batch	Temp	Additional Examples	Total Attempted	Above Threshold	Solved 100%
batch-0	0.7-1.65	0	47	21	0 (0.00%)
batch-1	0.7-1.65	0+data	50	21	1 (2.00%)
batch-2	0.7-1.25	2+data	48	20	1 (2.08%)
batch-3	0.7-1.35	4+data	45	21	0 (0.00%)

See Above Threshold definition.

4.4 Testing Introspective Verification Capabilities

At the conclusion of our experiments with Gemini Flash, we conducted an additional test to evaluate whether the model could recognize and reproduce the correct output when it was explicitly provided. Specifically, we supplied the true output for all puzzles to Gemini 4.4, accompanied by direct clues—effectively analogous to giving a student the correct answers labeled as "true_output." We also allowed the model up to three attempts per puzzle, providing its previous predictions and their corresponding scores after each attempt.

Despite these highly favorable conditions, Gemini achieved only a maximum accuracy of approximately 40%. This finding highlights an additional and noteworthy limitation. If the model correctly recognized that the input explicitly contained the true output, its success rate would reasonably be expected to approach 90% or higher. Conversely, if the model failed to detect

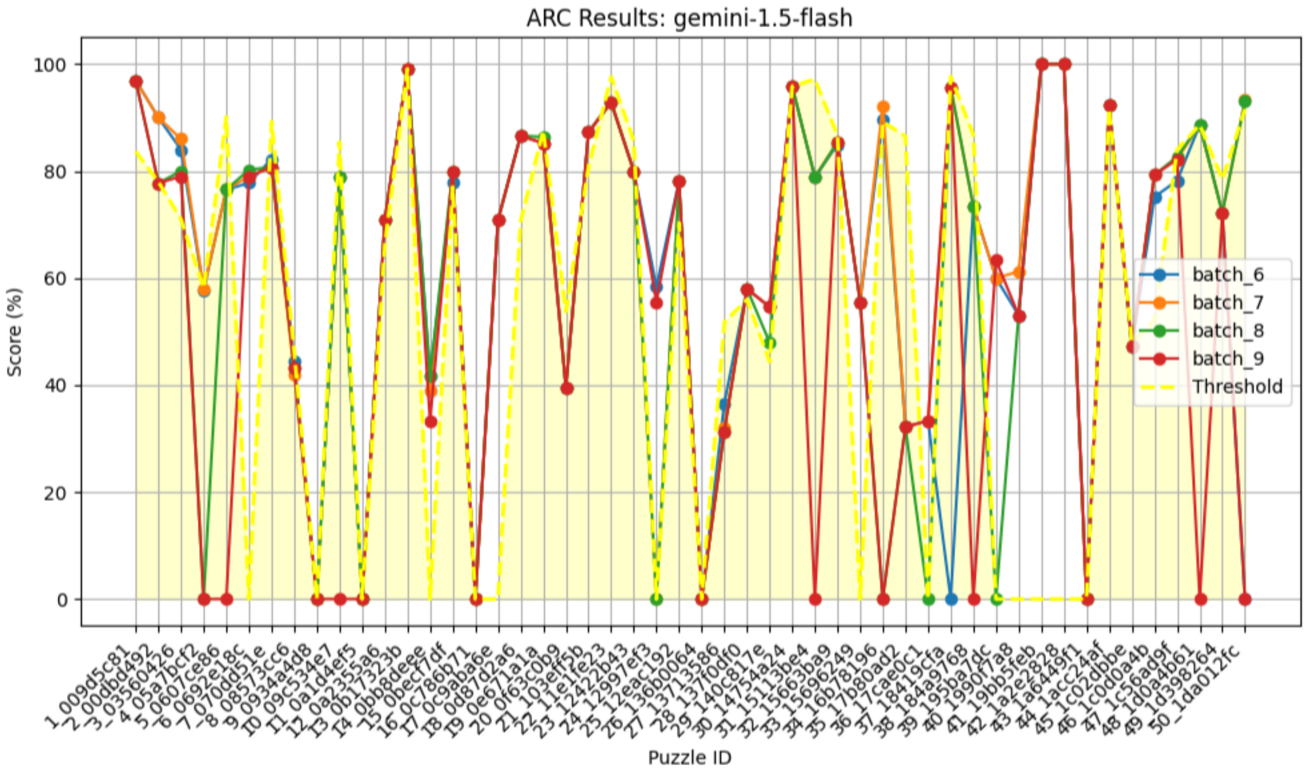

Figure 17. ARCAGI Puzzles Batch 6,7,8,9

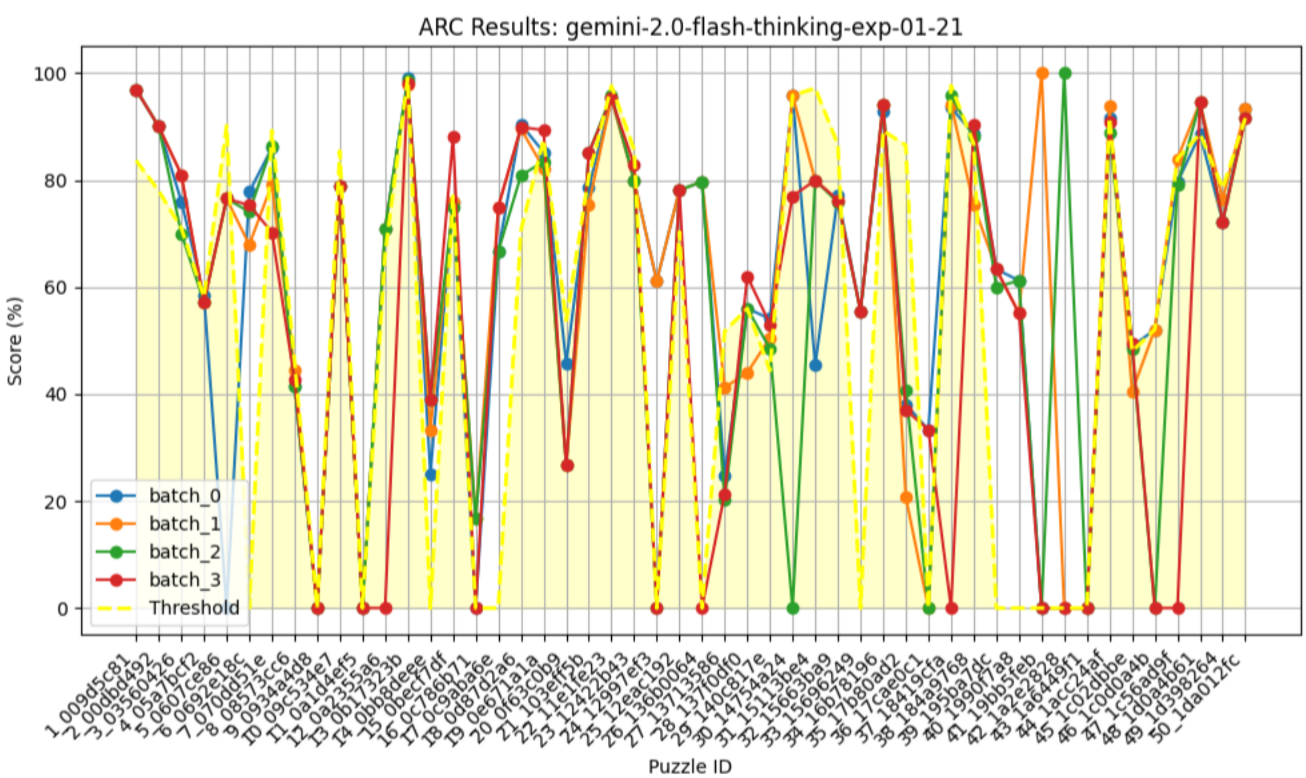

Figure 18. ARCAGI Puzzles Batch 0,1,2,3

the presence of the leaked true output, its accuracy would be expected to remain near 5%, corresponding to chance-level or uninformed attempts. The allowance of up to three attempts was intended to enable the model to infer—based on the outcomes of earlier predictions—that the provided true output was correct. However, the persistently low accuracy suggests that the model does not consistently exhibit mechanisms for consistency checking, introspective verification, and common-sense reasoning.

API Input Data with True Output


```
{
    "training": {
        "example_1": {
            "input": [........],
            "output": [........]
        },
        "example_2": {
            "input": [........],
            "output": [........]
        },
        "example_3": {
            "input": [........],
            "output": [........]
        },
    },
    "previous_attempts": {"Textual details of upto 5 previous attempts and best
       guesses in matrix format"},
    "test_input": [........],
    "true_output": [........],
}
```


5 Conclusion of Experiments

The experiments presented in this study were designed as targeted, diagnostic probes rather than as statistically exhaustive benchmarks. Within this limited but controlled scope, they reveal consistent patterns in how contemporary large language models (LLMs) respond to tasks that require grounded reasoning, self-verification, or adaptation beyond surface-level pattern completion.

Across multiple experimental settings—including prompt reformulation, few-shot conditioning, and chain-of-thought elicitation—we observed that improvements in output quality were often fragile and highly sensitive to prompt structure. While certain prompt strategies could mitigate specific failure cases, these effects were neither robust nor generalizable across tasks. Importantly, no experiment demonstrated a reliable emergence of new reasoning strategies or sustained correction of previously identified errors solely through inference-time intervention.

These observations support two empirically grounded conclusions. First, within the tested scenarios, inference-time prompting alone did not reliably extend the model's reasoning capabilities beyond patterns already supported by its training distribution. Apparent improvements were better characterized as selective activation of latent behaviors rather than as evidence of dynamic learning or strategy acquisition. This distinction is critical when interpreting claims of "learning" or "self-improvement" at inference time.

Second, the experiments highlight a recurring limitation related to self-assessment. In multiple tasks, models produced confident outputs despite being incorrect, and they showed limited capacity to internally recognize or flag such errors without explicit external constraints. While this does not imply that all forms of meta-cognition are absent from LLMs, it does indicate that, in the tested configurations, error awareness and self-auditing were not reliably engaged.

At a broader level, these findings do not falsify the possibility that scaling, architectural changes, or hybrid systems could yield more robust generalization in the future. However, they do caution against interpreting fluent language behavior and prompt-sensitive improvements as evidence of grounded intelligence or autonomous reasoning. The results instead suggest that current LLM capabilities, as evaluated here, remain tightly coupled to training-time representations and lack stable mechanisms for internal verification or adaptive restructuring.

Finally, the proposed RECAB framework should be interpreted as a conceptual proposal motivated by these diagnostic observations, rather than as a validated benchmark. Its inclusion is intended to stimulate discussion around evaluation criteria that emphasize self-auditing, constraint awareness, and adaptive behavior, rather than to replace existing benchmarks at this stage.

In summary, the experiments demonstrate specific and reproducible limitations in contemporary LLM behavior under targeted conditions. While they do not justify absolute claims about the impossibility of Artificial General Intelligence, they underscore a critical gap between surface-level linguistic competence and the forms of grounded, self-regulating intelligence typically associated with general reasoning systems. Addressing this gap will likely require advances that extend beyond prompt engineering and scale alone.

6 Evaluating Intelligence in Large Language Models: Observed Limitations and Open Criteria

Despite substantial progress driven by increased model scale and data availability, several fundamental limitations of contemporary large language models (LLMs) remain unresolved. Challenges such as robust reasoning under distributional shift, grounded semantic understanding, and stable alignment with human objectives appear to require advances beyond incremental scaling of existing architectures. Consequently, current LLMs operate most reliably as probabilistic inference systems augmented by human oversight rather than as autonomous, domain-general reasoning agents.

Empirical observations reported in the literature and corroborated by this study indicate recurring limitations in state-of-the-art models, including:

- Weak evidence of grounded or causal semantic representations
- Fragility in common-sense reasoning, particularly under novel or underspecified conditions
- Sensitivity to prompt formulation, often yielding shallow or inconsistent inference
- Disproportionately high computational and data requirements relative to task complexity
- Limited interpretability of internal representations and decision pathways
- Vulnerability to adversarial or strategically misleading inputs
- Generation of hallucinated or weakly justified outputs

A salient characteristic of these limitations is that they are predominantly revealed through external evaluation rather than endogenous model processes. Although LLMs can articulate known weaknesses when explicitly prompted[11,12], such responses are more plausibly attributable to learned statistical regularities in training data than to intrinsic mechanisms of self-monitoring or error detection. In practice, identification and mitigation of model failures remain dependent on human supervision or auxiliary verification systems.

As shown in Table **??**, leading models achieve strong performance on established benchmarks targeting common-sense and scientific reasoning. However, benchmark success does not consistently generalize to open-ended, distributionally novel, or real-world tasks. This discrepancy suggests a structural gap between benchmark-oriented performance metrics and the broader cognitive capacities typically associated with general intelligence.

In this context, the present work does not assert that existing models are incapable of developing more general reasoning abilities. Rather, it argues that prevailing evaluation methodologies insufficiently capture dimensions such as self-verification, adaptive reasoning under resource constraints, and calibrated uncertainty awareness. Motivated by this gap, we outline a resource-constrained evaluative perspective intended to foreground these dimensions. This framework is proposed as a conceptual direction for future research rather than as an empirically validated or comprehensive benchmark.

6.1 AGI Criteria Beyond Model Scaling

Recent evaluations indicate that state-of-the-art models such as Gemini and ChatGPT achieve strong performance on ARC-AGI-I while exhibiting markedly weaker results on ARC-AGI-II[31,32]. This divergence suggests that benchmark success does not necessarily imply robust abstraction or domain-general reasoning, but may instead reflect adaptation to specific data distributions, representational biases, or evaluation protocols.

The Abstraction and Reasoning Corpus (ARC)[3] remains one of the most stringent benchmarks for non-linguistic reasoning. However, like most contemporary evaluation frameworks, ARC primarily measures task-level outputs rather than the internal dynamics of reasoning and adaptation. Consequently, existing benchmarks provide limited insight into how models respond to novelty, uncertainty, or systematic failure. In contrast, human cognition is characterized by meta-cognitive monitoring, iterative hypothesis revision, and adaptive behavior under finite cognitive resources.

To articulate these underrepresented dimensions, we briefly propose the **Resource-Efficient Cognitive Autonomy Benchmark (RECAB)** as a conceptual research direction rather than an empirically validated benchmark. RECAB shifts the evaluative focus from static task accuracy toward process-level criteria governing adaptive cognition under resource constraints. At a high level, RECAB is structured around three interrelated capabilities:

- **Self-Audit:** detection of internal inconsistencies, errors, or performance limitations without external supervision.
- **Self-Directed Strategy Generation:** construction of alternative reasoning or problem-solving strategies in response to identified limitations.
- **Iterative Implementation:** execution and evaluation of selected strategies within fixed computational and memory budgets.

Within this framework, general intelligence is operationalized not merely as task competence but as sustained improvement under bounded resources, including the capacity for corrective unlearning and principled trade-offs between exploration, exploitation, and efficiency. This perspective departs from idealized models such as AIXI[39], Gödel Machines[38], and Recursive Self-Improvement paradigms[37], which assume unbounded or theoretically optimal computation. RECAB instead foregrounds finite-resource adaptation as a defining condition of practical intelligence.

We emphasize that RECAB is not intended as a definitive test of AGI nor as a replacement for existing benchmarks. Rather, it functions as a conceptual lens for identifying evaluation dimensions—meta-cognitive self-assessment, adaptive iteration, and resource-bounded autonomy—that remain weakly captured by prevailing task-centric methodologies. The formalization and empirical validation of these criteria are left as directions for future work.

7 Conclusion

This study examined the behavior of contemporary large language models (LLMs) through a series of targeted, diagnostic experiments designed to probe reasoning, generalization, and self-verification under constrained conditions. The results show that, while LLMs exhibit impressive linguistic fluency and can solve a wide range of benchmark-style tasks, their performance remains uneven when confronted with novel, underspecified, or practically grounded problems. These limitations are not uniform failures, but recurring patterns observed across multiple experimental settings discussed in this paper.

A central empirical observation is that LLM performance is strongly shaped by prompt structure and task formulation. Improvements achieved through rephrasing, few-shot prompting, or chain-of-thought elicitation were often fragile and task-specific, rather than robust or transferable. Within the scope of our experiments, such strategies did not reliably induce new reasoning capabilities or sustained correction of previously identified errors. This suggests that inference-time prompting primarily activates latent behaviors learned during training, rather than enabling genuine learning or adaptive reasoning at runtime.

The experiments also reveal a discrepancy between fluent textual reasoning and performance in tasks requiring grounded or cross-modal understanding. While models can generate coherent descriptions or explanations, this competence does not consistently transfer to visual, spatial, or mechanical reasoning scenarios. These findings align with broader concerns in the literature regarding the absence of stable, integrated representations that support reasoning across modalities.

Importantly, the results presented here do not establish that scaling model size or training data cannot yield further improvements. Rather, they caution against interpreting surface-level performance gains or benchmark success as evidence of general intelligence. The failure cases analyzed in this work indicate that increased scale alone does not reliably address challenges related to self-assessment, error awareness, or adaptive behavior under constraints—dimensions that are central to many definitions of general intelligence.

Within this context, we argue that current evaluation practices may overemphasize outcome-based task success while underrepresenting process-oriented aspects of cognition, such as how a system identifies uncertainty, manages limited resources, or responds to failure. The conceptual RECAB framework introduced in this paper is intended to highlight these gaps and to motivate future empirical research, rather than to serve as a validated benchmark or definitive alternative to existing evaluations.

In summary, the evidence presented supports a cautious interpretation of current LLM capabilities. Contemporary models are powerful and versatile tools, but their observed behavior suggests that fluent language generation alone should not be conflated with grounded reasoning or general intelligence. Progress toward more general and autonomous forms of AI will likely require advances that go beyond prompt engineering and parameter scaling, incorporating mechanisms for self-evaluation, adaptation, and operation under explicit constraints.

7.1 Key Takeaways

- LLM performance improvements achieved through prompting strategies are often fragile and task-specific, with limited evidence of robust transfer or inference-time learning.
- High benchmark scores do not consistently translate into reliable performance on novel, grounded, or cross-modal tasks.
- Current LLMs show limited capacity for self-assessment or error awareness without external guidance.
- Scaling model size and data alone does not reliably address these limitations, highlighting the need for complementary evaluation criteria and architectural approaches.

Glossary

Above Threshold Denotes the degree to which the model's predicted output matrix exceeds a predefined similarity benchmark relative to the true output matrix.. 17, 21

Post-hoc Corrections Adjustments applied to a model's outputs or behavior after observed failures, typically through fine-tuning, reinforcement learning, rule-based filtering, or prompt-level constraints.. 12

SotA SotA LLMs refers to State-of-the-Art Large Language Models such as ChatGPT, Gemini, DeepSeek, and Grok. These models represent the most advanced implementations of transformer-based neural architectures currently available.. 10

Threshold The minimum acceptable similarity score (measured as normalized pixel-wise correspondence) required for meaningful alignment between prediction and ground truth. For example, if the threshold for a given puzzle is **78%**, a model output achieving **83%** similarity is recorded as **5% above threshold**. This metric applies only to puzzles where the **input and output matrices share identical dimensions**; for all other cases involving dimension changes or structural transformations, the threshold value is set to **zero**, as direct element-wise comparison is not applicable.. 17

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants, animals, or sensitive data requiring ethical approval.

Consent for publication

Not applicable. No individual person's data or identifiable information is included in this manuscript.

Availability of data and materials

Additional data and resources will be made available upon reasonable request.

Competing interests

The authors declare that there are no competing interests.

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Authors' contributions

Reshid Mehmood has worked on the proposed scheme, visualization, Dr.Eid Rehman on validation, and Dr.Muhammad Habib on analysis.

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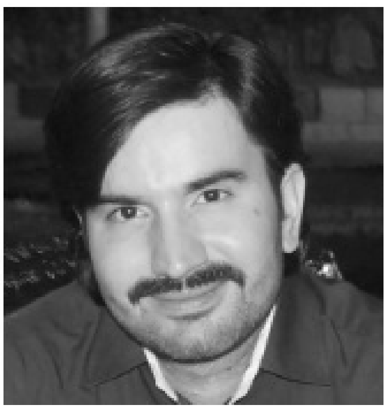
Dr. Rehman's research work bridges theory and practical application, contributing valuable insights to cutting-edge areas critical to today's technological advancements. He remains actively engaged in research, mentoring students, and participating in collaborative projects to foster innovation and excellence in Computer Science.
His commitment to academic excellence and research innovation continues to inspire the next generation of computer scientists at the University of Mianwali and beyond.



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